



Tethys Platform™



Web Application Development for Hydrologic Modeling and Visualization

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TETHYS GEOSCIENCE™ FOUNDATION

Mission Statement: “The Tethys Geoscience Foundation strives to enhance the impact of **geoscience for the betterment of society and the planet**. We support the **development of innovative tools** that; lead to **informed decision-making** in response to natural disasters, contribute to the creation of sustainable environmental policies, and provide improved management of Earth’s resources.”

<https://tethysgeoscience.org/>



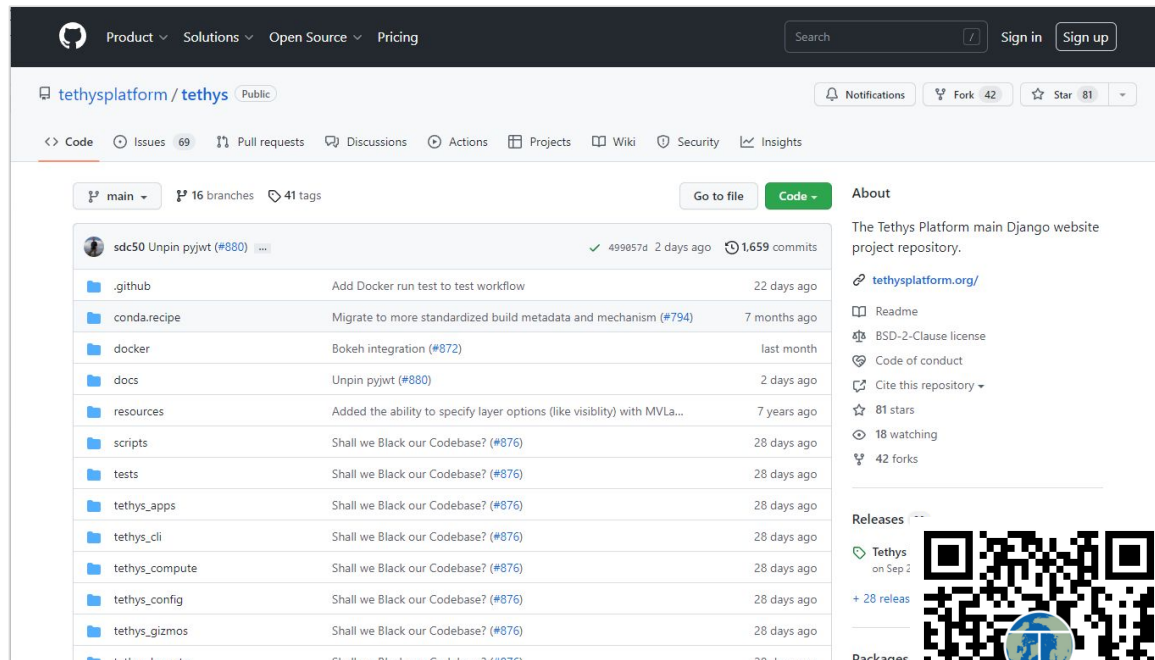
Tethys Platform





Tethys Platform

- Open Source
- Easy as 1-2-3!
- Tethys Portal
- Interoperable GIS
- Tethys Apps
 - Standard
 - Tethys Dash
 - Component



<https://github.com/tethysplatform>





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If you already have conda...

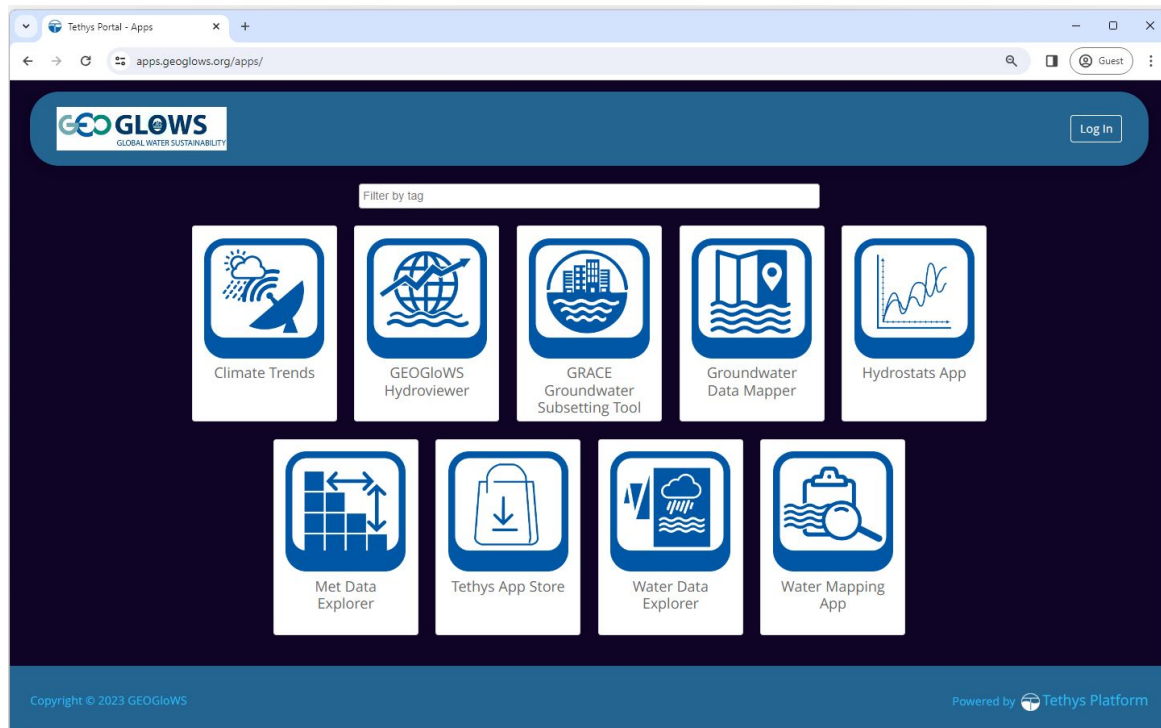
```
conda create -n tethys -c conda-forge tethys-platform  
conda activate tethys  
tethys quickstart
```

<https://docs.tethysplatform.org>



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“Big Box of GIS Stuff” -Dan Ames, BYU





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Tethys SDK: Languages/Libraries

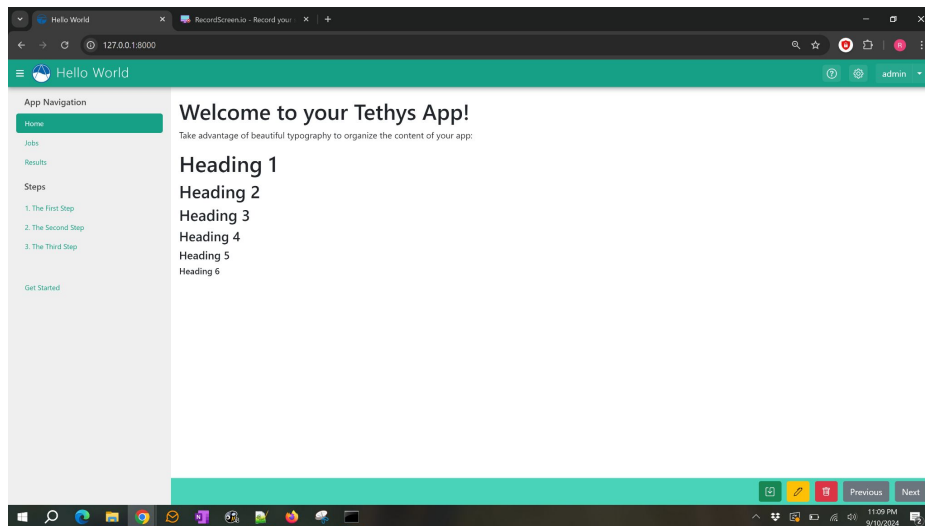




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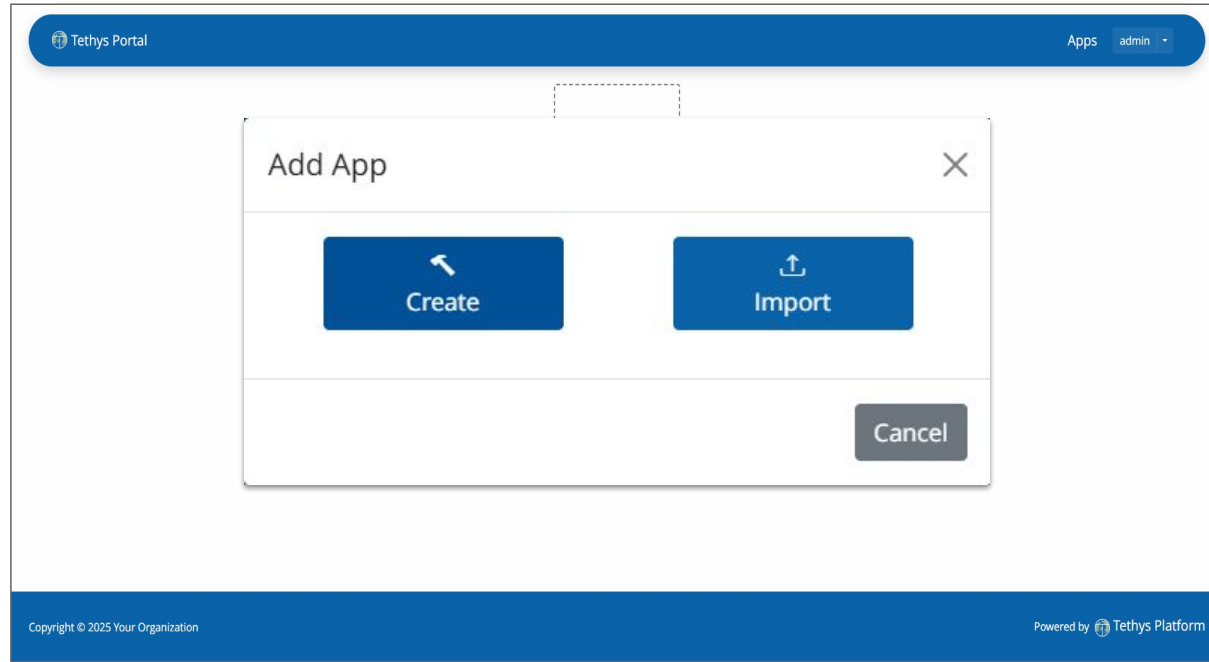
```
tethys scaffold my_app_name  
cd tethysapp-my_app_name  
tethys install -d
```





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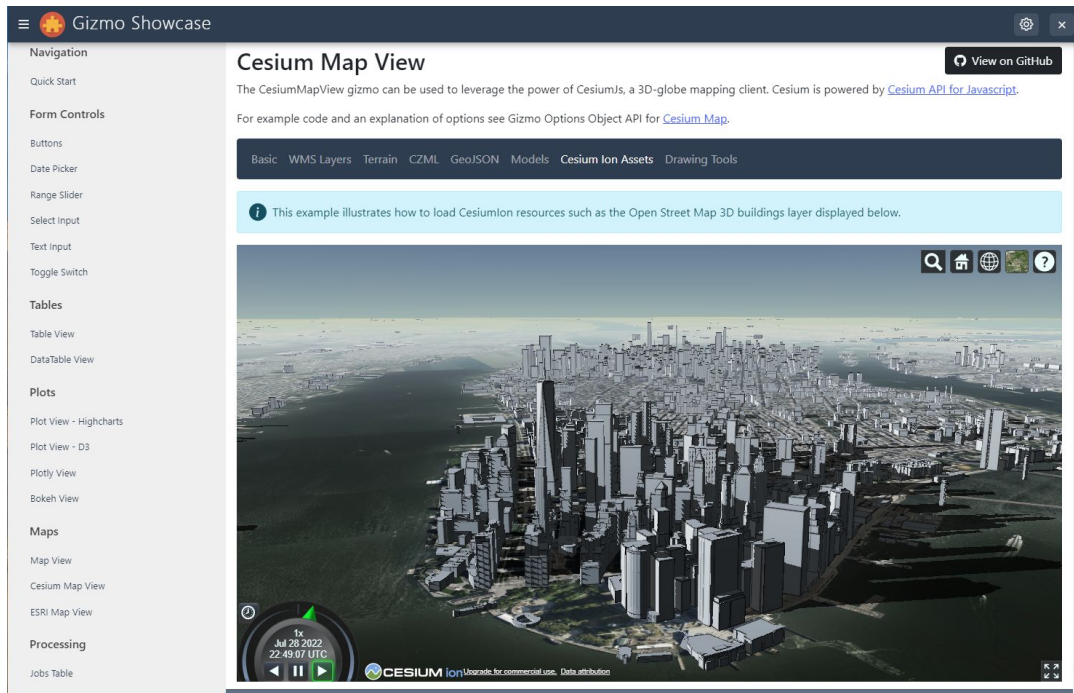


Coming soon: Scaffold through Portal!



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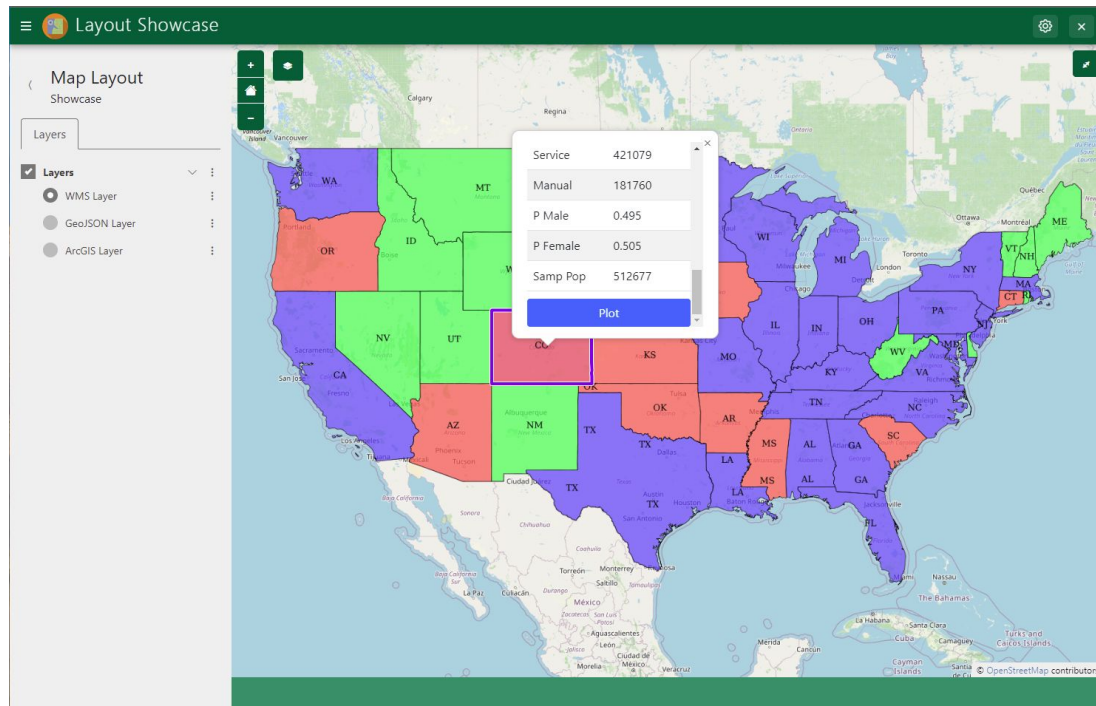


See: <https://demo.tethysgeoscience.org/apps/gizmo-showcase/>



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See: <https://demo.tethysgeoscience.org/apps/layout-showcase/>



Tethys Platform

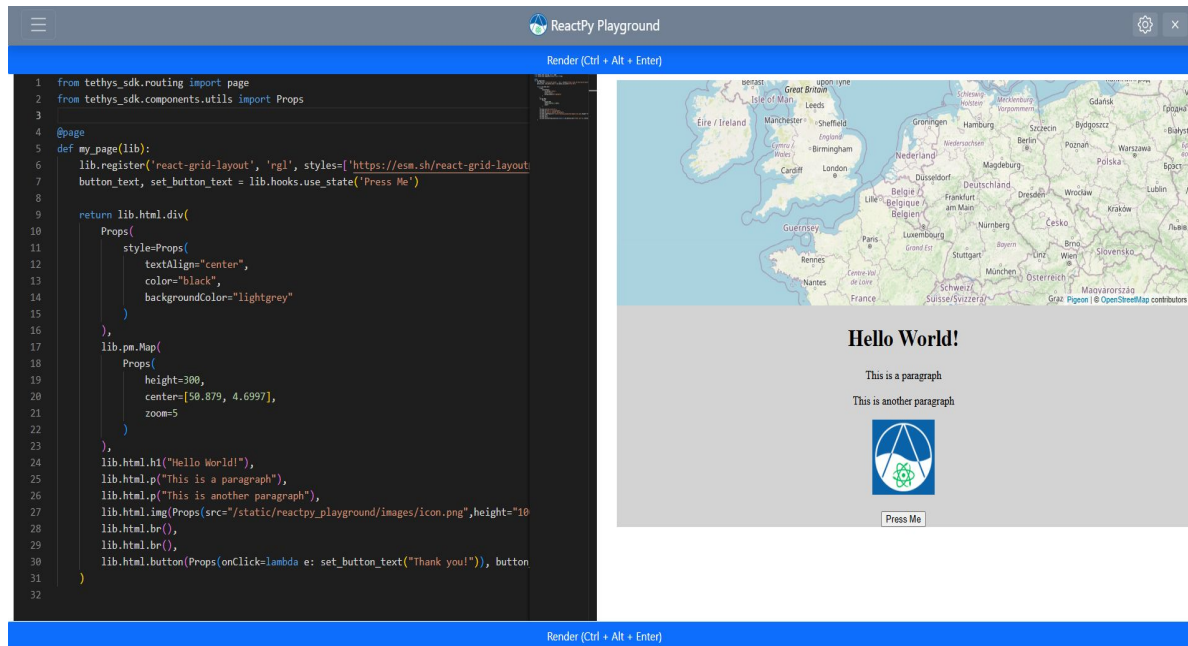
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Component Apps





What is a Tethys Component App?



A framework for building web applications using reactive, Python-based components.

- Built on ReactPy and Django
- Edit a single Python file
- Don't directly write any HTML/CSS/JavaScript
- Write much less code



What is a Component?

Reusable, independent code blocks (i.e. a function) that define the structure and behavior of the User Interface (UI). They accept inputs (props or properties) and return elements that describe what should appear on the screen.





Component App Resources



[Tethys Platform Quick Start Documentation](#)

[Component Playground App Code](#)

[Component App Basic Tutorial](#)

[Tethys Component App Core Documentation](#)

[Tethys Custom Components API](#)

[Tethys Component Utils API](#)



DEMO





Tethys Dash





What is TethysDash?



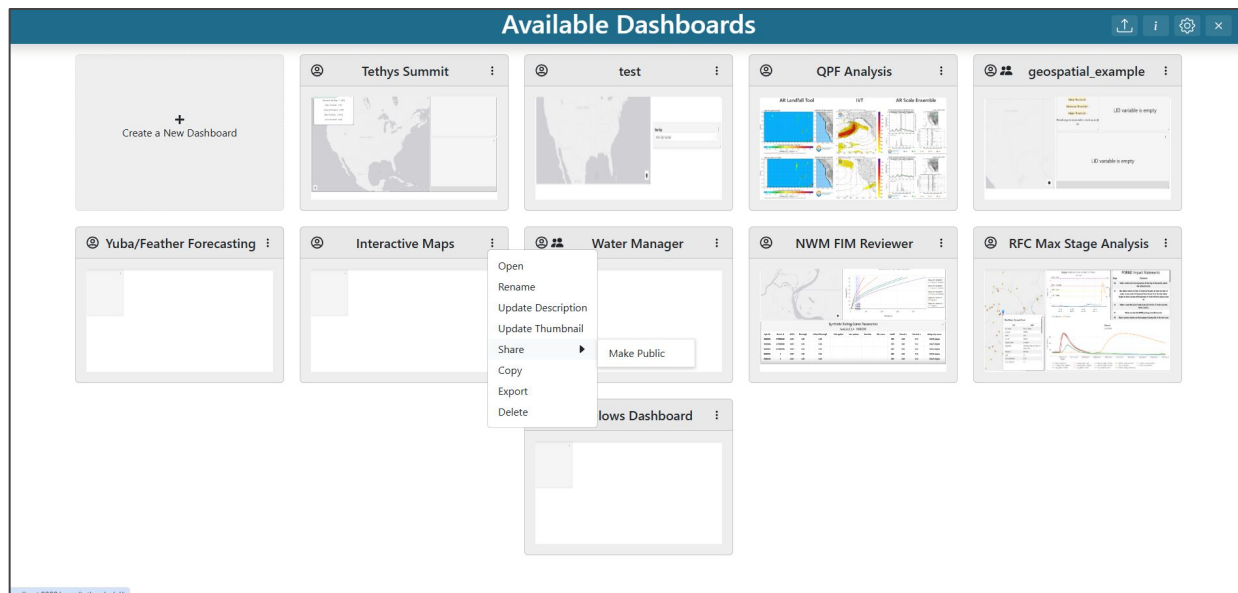
- A framework for building no-code/low-code dashboards and web applications.
 - Intuitive, GUI-based dashboard and app builder
 - Highly customizable layouts to suit diverse needs
 - Flexible plugin architecture for adding rich visualizations
 - Easily share, import, and export dashboards and apps



Dashboard Landing Page



- Create, edit, and delete user owned dashboards
- Share dashboards in the tethys portal for others to see
- Export dashboards into JSON file for sharing/altering
- Import dashboards from JSON files

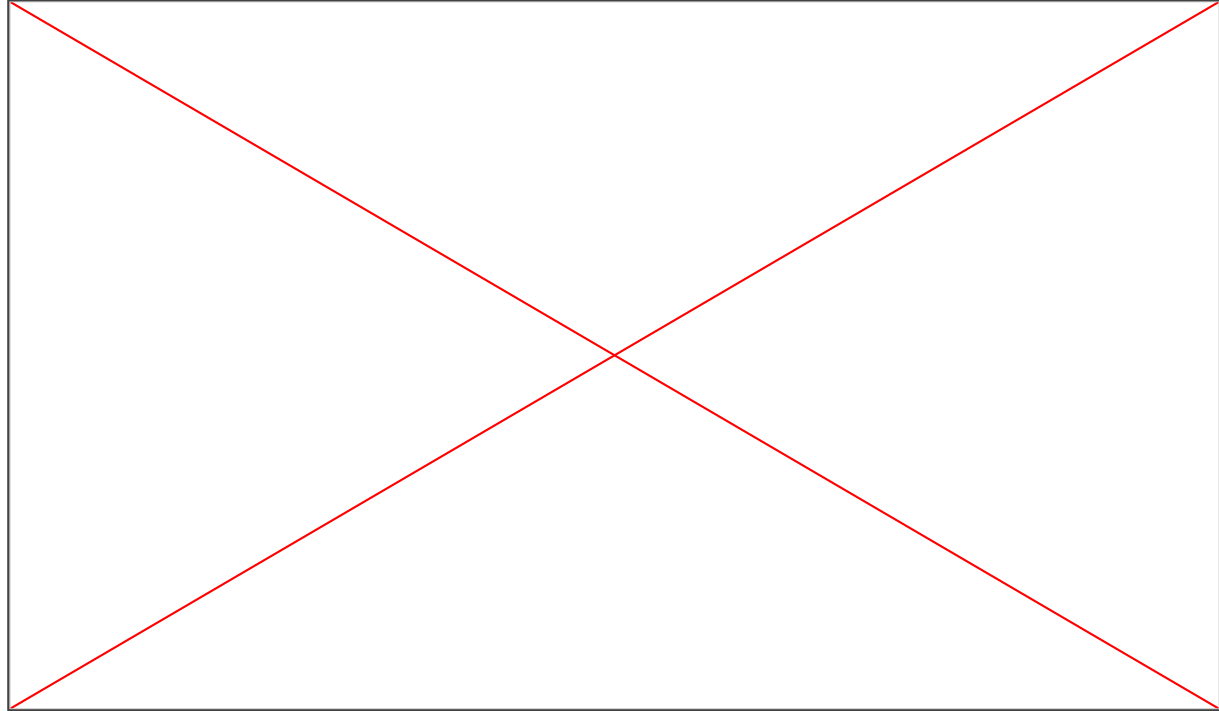




Dashboards

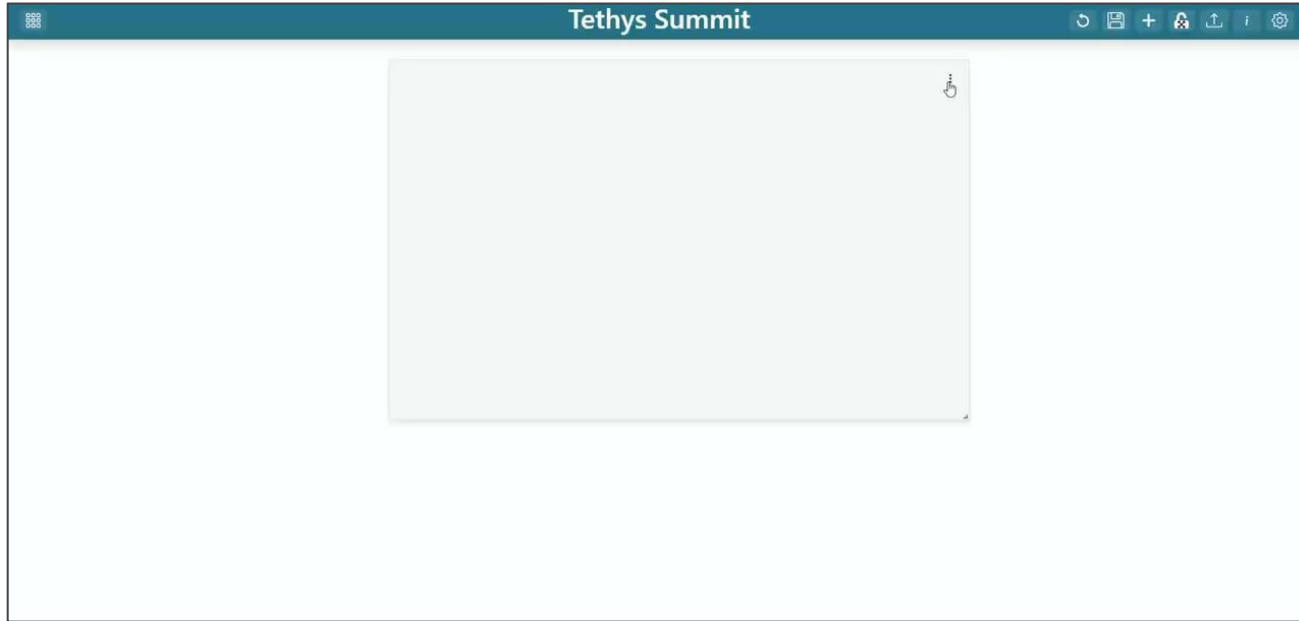


- Dashboards are composed of individual items
- Dashboard items can be added, moved, resized, or deleted from the interface





Visualizing Data



- Dashboard items can be edited to show configurable visualizations
- Default visualizations
 - Maps
 - Custom Images
 - Text
 - Variable Inputs



Creating Dynamic Components - Variable Input



- Variable inputs, similar to variables in programming, are values that are stored and represented as a string.
- Variables inputs can be created and updated through the **Variable Input** visualization type or as a field in a map layer.
- References work similar to javascript, with the syntax being **`${<some variable>}`**

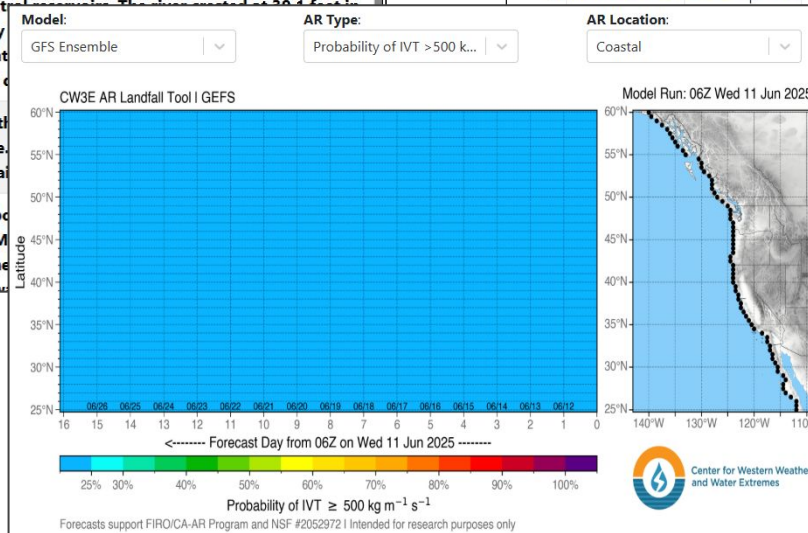
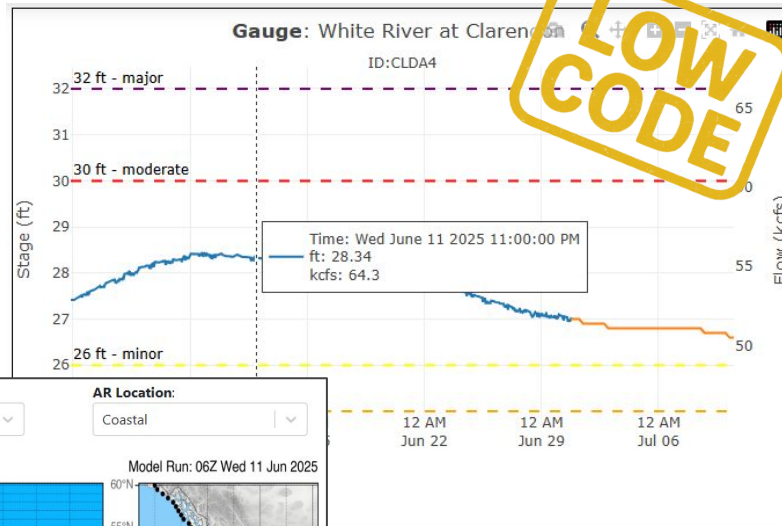


What if I want more?

CLDA4 Impact Statements

Stage	Statement
43	Similar flood contions to flood of record, which occurred prior to the constuction of the flood control reservoirs, was 43.3 feet in April 1927. Evacuations will be necessary.
39	River has not been this high since completion of flood control. The highest crest since completion of flood control was 35 feet in May 1973. River Valley is like a large lake. Most cultivated land in floodplain affected by roads near the river in Prairie, Monroe, and Arkansas count also be flooded. Driving at night is especially d
35	River flooding is very serious. Second highest crest since t was 35 feet in May 1973. River Valley is like a large lake. cropland in the floodplai
33	Most all of the land within the levees is inundated. Large po Wildlife Refuges are flooded. River crested at 33 feet in M May 1973. High Mississippi River worsens flooding on the Clarendon to Saint Charles will be impacted by the high w

Tables

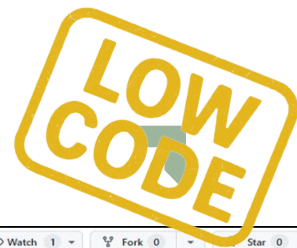


Active Charts

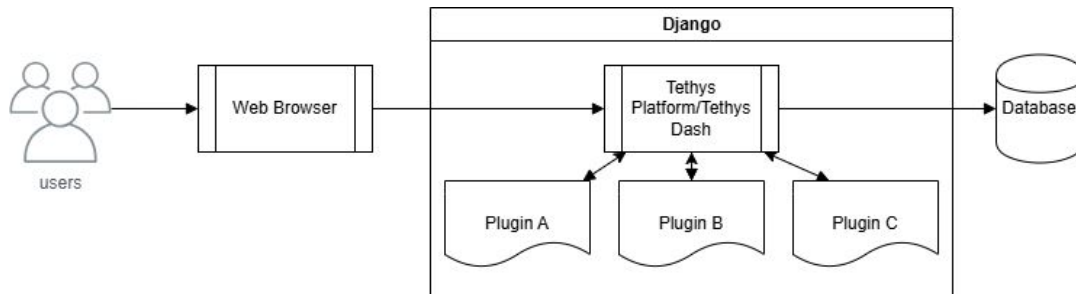
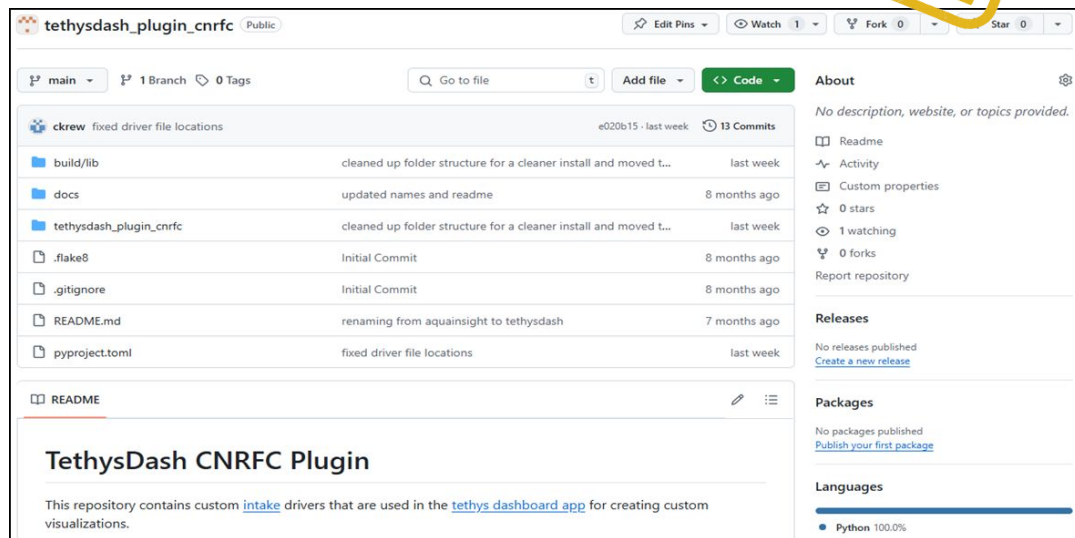
Visualizations with specific inputs/arguments



Creating Custom Visualizations



- TethysDash allows developers to create and install visualization plugins
- Python based using the Intake package





Plugin Example

- Available Visualization Types
 - Plotly Charts
 - Table
 - Image
 - Card
 - Text
 - Variable Input
 - Map
 - Map Layer
- Read method will run for visualization processing and return the necessary data for the specified visualization type.

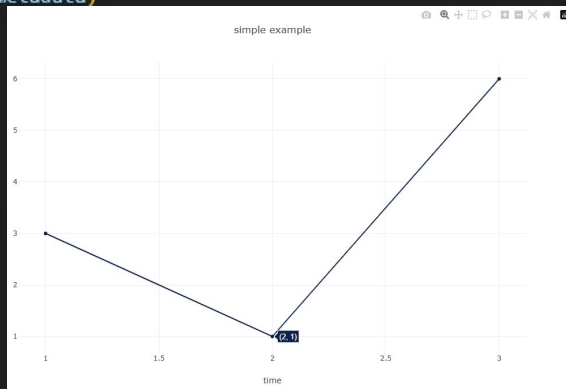
```
from intake.source import base
```

```
class PlotlyExample(base.DataSource):  
    container = "python"  
    version = "0.0.1"  
    name = "plotly_example"  
    visualization_args = {} # Arguments provided from TethysDash used for processing  
    visualization_group = "Example" # Group for visualization plugin discovery  
    visualization_label = "Plotly Example" # Visualization plugin name  
    visualization_type = "plotly" # Type of visualization  
    visualization_tags = [  
        "example",  
        "plotly",  
    ] # Tags for visualization plugin discovery  
    visualization_description = "An example plugin for the plotly visualization"
```

```
def __init__(self, metadata=None):  
    super().__init__(metadata=metadata)
```

```
def read(self):  
    """  
    Return plotly information  
    """  
    data = [  
        {  
            "type": "scatter",  
            "x": [1, 2, 3],  
            "y": [3, 1, 6],  
        },  
    ]
```

```
    layout = {  
        "title": "simple example",  
    }  
    return {"data": data, "layout": layout}
```



LOW
CODE



DEMO

